

The slope formula



- 1
- | | | | |
|-----------------|------|------|------|
| a 3 | b -3 | c -1 | d -1 |
| e $\frac{3}{5}$ | f -1 | g 0 | h 0 |
- 2 No. Reversing the order does not affect the final outcome of the slope. As long as you maintain the correct order by subtracting the corresponding y and x coordinates, the slope should be the same.
- 3
- | | | | |
|-------------|-------------|-------------|-----------|
| a Undefined | b Undefined | c Defined | d Defined |
| e Undefined | f Defined | g Undefined | h Defined |
- 4
- | | |
|------------|-----|
| a $y = 25$ | b 5 |
|------------|-----|
- 5
- | | | | |
|-------------|------------|------------|-----------------------|
| a $t = 3$ | b $t = 15$ | c $d = 20$ | d $c = -\frac{12}{7}$ |
| e $t = -23$ | | | |
- 6 890 m
- 7
- | | | |
|-----|-----|------|
| a 0 | b 4 | c No |
|-----|-----|------|
- 8
- | | | |
|-------------|------|-----|
| a undefined | b No | c 1 |
|-------------|------|-----|
- 9
- | | | | |
|-----------|-------------|-----|-----------|
| a Neither | b Undefined | c 0 | d Neither |
| e Neither | f Neither | g 0 | |
- 10
- | | |
|---|---|
| a The change in x is equal to zero. The line is vertical. | b The line is horizontal. The change in y is equal to zero. |
|---|---|



- 11
- a No
 - b Yes
 - c No

12 D

- 13
- a 0
 - b Undefined

14 B

15 -20

Additional questions

- 16
- a
 - i 14
 - ii 7
 - iii 2
 - b
 - i 12
 - ii 4
 - iii 3
 - c
 - i -12
 - ii 4
 - iii -3
 - d
 - i -6
 - ii 2
 - iii -3

- 17 Because a vertical line has no run. So the formula for the slope $\frac{\text{rise}}{\text{run}}$ will have a denominator of zero, and any fraction with a zero denominator is undefined.

18 0

- 19
- a 2
 - b $-\frac{7}{5}$
 - c $\frac{1}{2}$
 - d $-\frac{2}{3}$

20

a -2



b As x increases, the value of y decreases.

21

a 3

b $\frac{3}{5}$

c -4

d -3

e 0

f 0

g 1

h -1

22

a $\frac{5}{3}$

b $\frac{5}{3}$

c Yes

23 $m = 1$